Assignment_3_Data_Analysis

2026-03-01

```
library(readr)
library(jsonlite)
library(tidyverse)
library(readxl)
library(ggplot2)
library(tidyr)
library(dplyr)
```

```
json<- "basic-questionnaire-study_all-4.json"
data <- fromJSON(json, simplifyVector = FALSE)
#glimpse(data), this function has been commented out as to not overwhelm the Markdown
```

```
rows <- lapply(data, function(p) {
  answers <- p$answers
  name <- trimws(answers[["first-question-set_1"]]$answer[["q1-name"]] %||% "")
  get_answer <- function(key) {
    ans <- answers[[key]]$answer
    if (length(ans) > 0) unlist(ans)[[1]] else NA
  }
  list(
    Name        = name,
    Base_Chart  = get_answer("vis_1_2"),
    Chart_3D    = get_answer("vis_2_3"),
    Hole_Chart  = get_answer("vis_3_4")
  )
})
`%||%` <- function(a, b) if (!is.null(a)) a else b
data_before_cleaning <- do.call(rbind, lapply(rows, as.data.frame))
print(data_before_cleaning)
```

```
##                 Name Base_Chart Chart_3D Hole_Chart
## 1                Nic      15.00    40.00      15.00
## 2             Hayden       0.00     0.00       0.00
## 3              Kenny      16.67    12.50      25.00
## 4              Katie      17.00    10.00      20.00
## 5             Connor      50.00    40.00      35.00
## 6              Paola      10.00     5.00       3.00
## 7               Toby      20.00     7.00      17.00
## 8              Aidan      20.00    20.00      20.00
```

```
## 9              stacy      15.00    25.00      10.00
## 10             Emily      20.00     7.00      15.00
## 11   Josh Bornfriend      50.00    25.00     100.00
## 12             Tyler       0.33     0.33       0.25
## 13         Annabelle      50.00    20.00      75.00
## 14             Delia      15.00    20.00      30.00
## 15                 -      20.00    20.00      20.00
## 16               Kai      70.00    65.00      75.00
## 17           Zackary      25.00    15.00      17.00
## 18            Walter      10.00    10.00      15.00
## 19               Lia      15.00    20.00      10.00
## 20           Abigail      40.00    30.00      10.00
## 21      Sarah Bodkin      10.00    15.00      10.00
## 22              Izzy       0.00     0.00       0.00
## 23           Braedan      50.00    25.00      33.00
## 24             Megan      15.00    10.00      25.00
## 25             David       0.00     0.00       0.00
## 26             Jesse      30.00    10.00      30.00
## 27           Darlina      20.00    15.00      14.00
## 28          OkaNdlox       5.00     3.00      13.00
## 29              Test      10.00    20.00      35.00
## 30             Gavin      20.00    30.00       0.00
## 31 Mikayla (Testing)      25.00    25.00      25.00
## 32                           NA       NA         NA
## 33                           NA       NA         NA
## 34                           NA       NA         NA
## 35                           NA       NA         NA
## 36                           NA       NA         NA
## 37                           NA       NA         NA
## 38                           NA       NA         NA
## 39                           NA       NA         NA
## 40                           NA       NA         NA
## 41                           NA       NA         NA
## 42                           NA       NA         NA
## 43                           NA       NA         NA
## 44                           NA       NA         NA
## 45                           NA       NA         NA
## 46                 sd        NA       NA         NA
## 47   Josh Bornfriend         NA       NA         NA
## 48                           NA       NA         NA
## 49            Emilia       0.00       NA         NA
## 50                           NA       NA         NA
## 51            Emilia      33.00       NA         NA
## 52                           NA       NA         NA
## 53              Emma         NA       NA         NA
## 54                           NA       NA         NA
## 55                           NA       NA         NA
```

```
## 56                                NA        NA          NA
## 57                                NA        NA          NA
## 58                                NA        NA          NA
## 59                                NA        NA          NA
## 60                                NA        NA          NA
## 61                                NA        NA          NA
## 62                                NA        NA          NA
## 63                                NA        NA          NA
## 64                                NA        NA          NA
## 65                                NA        NA          NA
## 66                                NA        NA          NA
## 67                                NA        NA          NA
## 68                                NA        NA          NA
## 69                                NA        NA          NA
## 70                                NA        NA          NA
## 71                                NA        NA          NA
## 72                                NA        NA          NA
## 73                                NA        NA          NA
## 74                                NA        NA          NA
## 75                                NA        NA          NA
## 76                                NA        NA          NA
```

```
data_final <- data_before_cleaning[-c(29, 31:76), ]
data_final[12, c("Base_Chart", "Chart_3D", "Hole_Chart")] <- data_final[12, c("Base_C
hart", "Chart_3D", "Hole_Chart")] * 100
data_final[, c("Base_Chart", "Chart_3D", "Hole_Chart")] <- round(data_final[, c("Base
_Chart", "Chart_3D", "Hole_Chart")])
data_final <- data_final %>%
  mutate(Name = paste0("P", row_number()))
print(data_final)
```

```
##    Name Base_Chart Chart_3D Hole_Chart
## 1    P1         15       40         15
## 2    P2          0        0          0
## 3    P3         17       12         25
## 4    P4         17       10         20
## 5    P5         50       40         35
## 6    P6         10        5          3
## 7    P7         20        7         17
## 8    P8         20       20         20
## 9    P9         15       25         10
## 10  P10         20        7         15
## 11  P11         50       25        100
## 12  P12         33       33         25
## 13  P13         50       20         75
## 14  P14         15       20         30
## 15  P15         20       20         20
## 16  P16         70       65         75
## 17  P17         25       15         17
## 18  P18         10       10         15
## 19  P19         15       20         10
## 20  P20         40       30         10
## 21  P21         10       15         10
## 22  P22          0        0          0
## 23  P23         50       25         33
## 24  P24         15       10         25
## 25  P25          0        0          0
## 26  P26         30       10         30
## 27  P27         20       15         14
## 28  P28          5        3         13
## 30  P29         20       30          0
```

```
true_value <- 20
```

```
data_final$Base_Chart_AbsError <- abs(data_final$Base_Chart - true_value)
data_final$Chart_3D_AbsError   <- abs(data_final$Chart_3D   - true_value)
data_final$Hole_Chart_AbsError <- abs(data_final$Hole_Chart  - true_value)

cat("Base Chart avg abs error =", mean(data_final$Base_Chart_AbsError, na.rm=TRUE), "
\n")
```

```
## Base Chart avg abs error = 12.2069
```

```
cat("3D Chart avg abs error =",   mean(data_final$Chart_3D_AbsError,   na.rm=TRUE), "
\n")
```

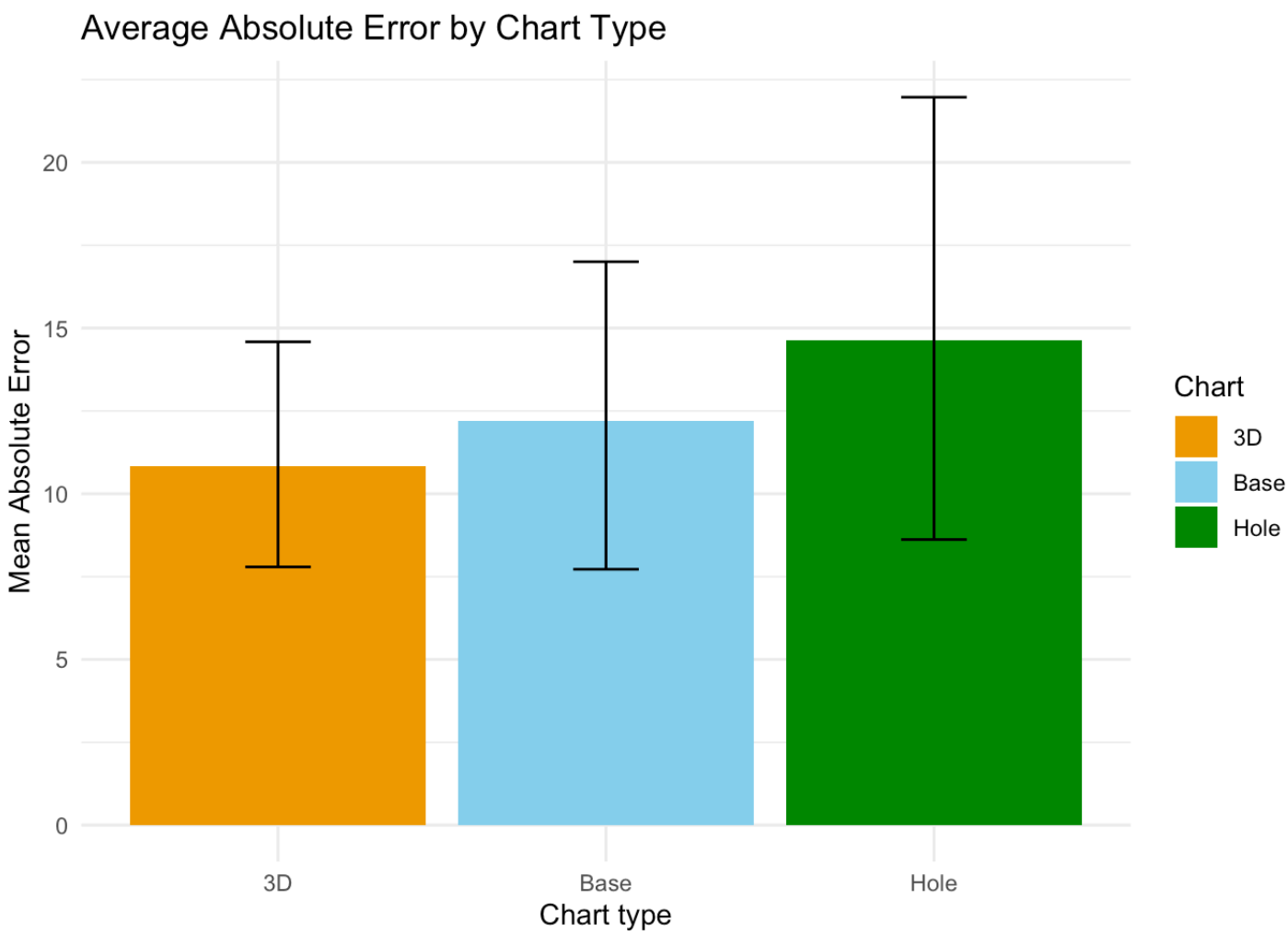
```
## 3D Chart avg abs error = 10.82759
```

```
cat("Hole Chart avg abs error =", mean(data_final$Hole_Chart_AbsError, na.rm=TRUE), "
\n")
```

```
## Hole Chart avg abs error = 14.62069
```

```
data_final_long_abs <- data_final %>%
  select(Name, Base_Chart_AbsError, Chart_3D_AbsError, Hole_Chart_AbsError) %>%
  pivot_longer(cols = -Name,
               names_to = "Chart",
               values_to = "Error") %>%
  mutate(Chart = recode(Chart,
                        "Base_Chart_AbsError" = "Base",
                        "Chart_3D_AbsError"   = "3D",
                        "Hole_Chart_AbsError" = "Hole"
  ))

ggplot(data_final_long_abs, aes(x = reorder(Chart, Error), y = Error, fill = Chart))
+
  stat_summary(fun = mean, geom = "bar") +
  stat_summary(fun.data = mean_cl_boot, geom = "errorbar", width = 0.2) +
  scale_fill_manual(values = c("Base" = "skyblue", "3D" = "orange2", "Hole" = "green
4")) +
  labs(
    title = "Average Absolute Error by Chart Type",
    x = "Chart type",
    y = "Mean Absolute Error"
  ) +
  theme_minimal()
```




```
cm_error <- function(reported, true) {
  ifelse(reported == true, 0,
         log2(abs(reported - true) + 1/8) - log2(1/8)
  )
}
data_final$Base_Chart_Error <- cm_error(data_final$Base_Chart, true_value)
data_final$Chart_3D_Error   <- cm_error(data_final$Chart_3D,   true_value)
data_final$Hole_Chart_Error <- cm_error(data_final$Hole_Chart,  true_value)
cat("Base Chart avg error:", mean(data_final$Base_Chart_Error, na.rm=TRUE), "\n")
```

```
## Base Chart avg error: 5.174076
```

```
cat("3D Chart avg error:",   mean(data_final$Chart_3D_Error,   na.rm=TRUE), "\n")
```

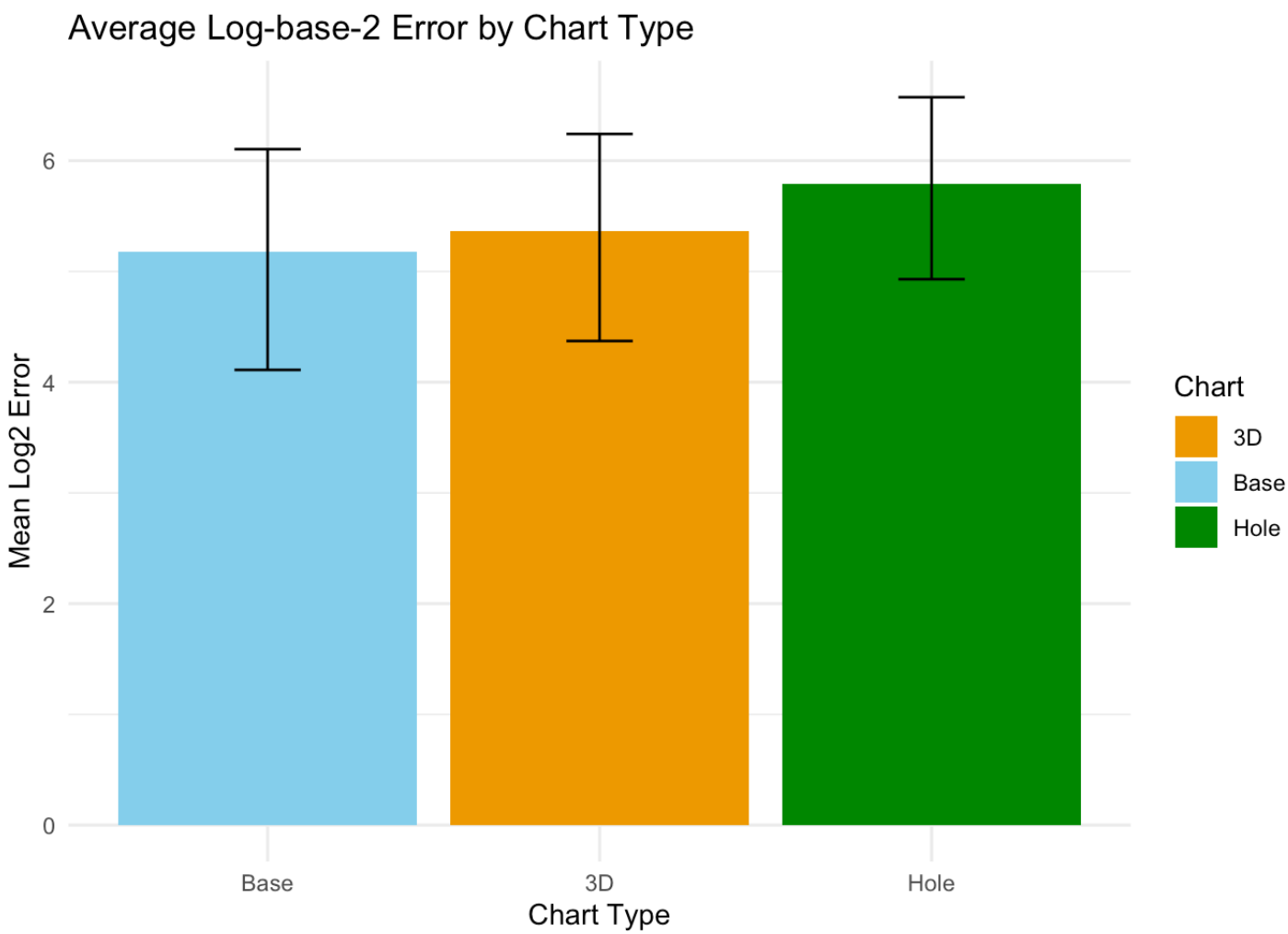
```
## 3D Chart avg error: 5.36265
```

```
cat("Hole Chart avg error:", mean(data_final$Hole_Chart_Error, na.rm=TRUE), "\n")
```

```
## Hole Chart avg error: 5.788551
```

```
data_final_long <- data_final %>%
  select(Name, Base_Chart_Error, Chart_3D_Error, Hole_Chart_Error) %>%
  pivot_longer(cols = -Name,
               names_to = "Chart",
               values_to = "Error") %>%
  mutate(Chart = recode(Chart,
                        "Base_Chart_Error" = "Base",
                        "Chart_3D_Error"   = "3D",
                        "Hole_Chart_Error" = "Hole"
  ))

ggplot(data_final_long, aes(x = reorder(Chart, Error), y = Error, fill = Chart)) +
  stat_summary(fun = mean, geom = "bar") +
  stat_summary(fun.data = mean_cl_boot, geom = "errorbar", width = 0.2) +
  scale_fill_manual(values = c("Base" = "skyblue", "3D" = "orange2", "Hole" = "green
4")) +
  labs(
    title = "Average Log-base-2 Error by Chart Type",
    x = "Chart Type",
    y = "Mean Log2 Error"
  ) +
  theme_minimal()
```



Beyond the scope of the project

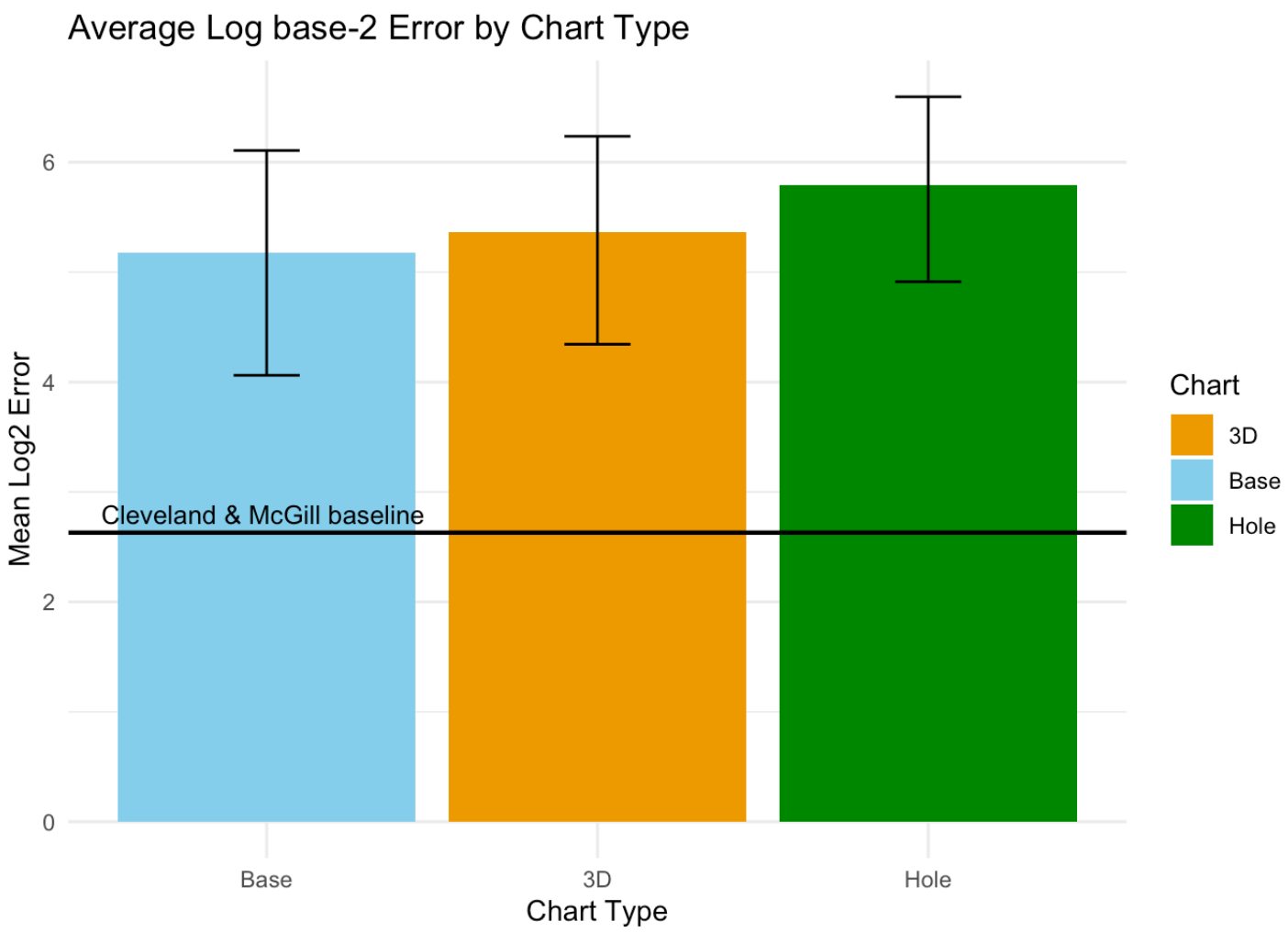
```
anova_result <- aov(Error ~ Chart, data = data_final_long)
summary(anova_result)
```

```
##             Df Sum Sq Mean Sq F value Pr(>F)
## Chart        2    5.7   2.874   0.421  0.658
## Residuals   84  572.9   6.820
```

```
TukeyHSD(anova_result)
```

```
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = Error ~ Chart, data = data_final_long)
##
## $Chart
##                 diff       lwr      upr     p adj
## Base-3D   -0.1885736 -1.824943 1.447796 0.9592023
## Hole-3D    0.4259011 -1.210468 2.062270 0.8090435
## Hole-Base  0.6144747 -1.021895 2.250844 0.6443526
```

```
ggplot(data_final_long, aes(x = reorder(Chart, Error), y = Error, fill = Chart)) +
  stat_summary(fun = mean, geom = "bar") +
  stat_summary(fun.data = mean_cl_boot, geom = "errorbar", width = 0.2) +
  geom_hline(yintercept = 2.63, color = "black", linewidth = 0.8) +
  annotate("text", x = 0.5, y = 2.8, label = "Cleveland & McGill baseline",
           hjust = 0, color = "black", size = 3.5) +
  scale_fill_manual(values = c("Base" = "skyblue", "3D" = "orange2", "Hole" = "green
4")) +
  labs(
    title = "Average Log base-2 Error by Chart Type",
    x = "Chart Type",
    y = "Mean Log2 Error"
  ) +
  theme_minimal()
```


Average Log base-2 Error by Chart Type
Mean Log2 Error
6
4
2
0
Cleveland & McGill baseline
Base
3D
Hole
Chart Type
Chart
3D
Base
Hole